Aegis-SOC: CI/CD, Security Assurance and Resource Consumption

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Aegis-SOC — CI/CD, Security Assurance and Resource Consumption Report

Companion document to `REPORT.md` (the architecture and evaluation report) Project: M.Tech (3rd Semester) Advanced Cloud Computing **Author:** Rudra Pratap **Date:** 19 October 2026 **Scope of this document:** delivery pipeline, security assurance activities, and measured resource consumption. Every figure below was produced by a command that is recorded in §8.

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1. Purpose and honesty statement

The architecture report answers *"does the platform detect and triage attacks?"*. This document answers two different questions that a reviewer or employer will ask next:

1. **“Can you ship it safely?”** — what automated gates exist, what did they find, and what was actually fixed.
2. **“What does it consume?”** — CPU, memory, latency and throughput under real concurrent load, since the free tier is metered on exactly those quantities.

Two claims are deliberately **not** made, because the verification did not complete in this environment:

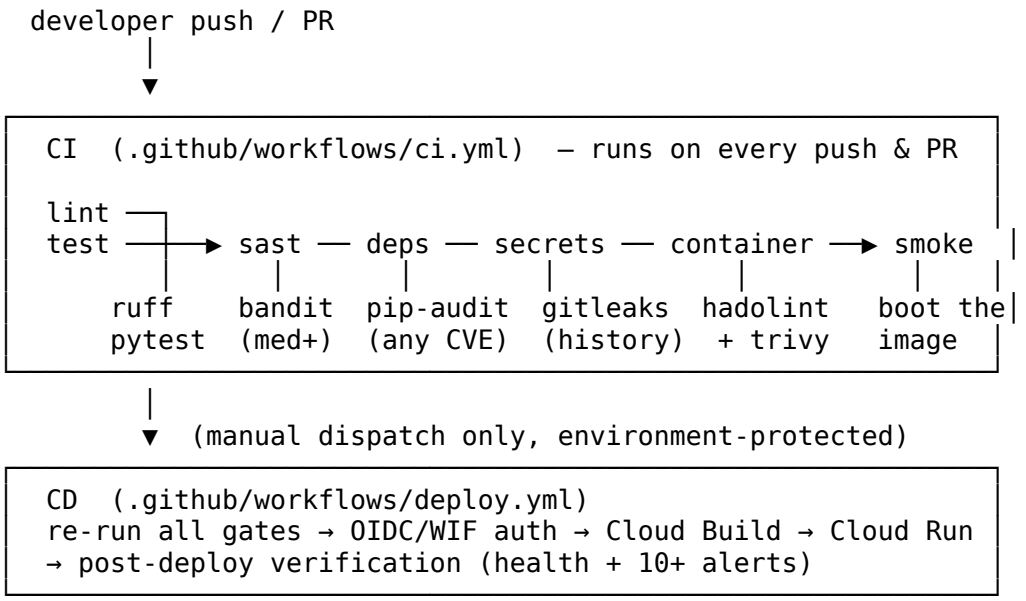
Not claimed	Why
“Image is fully patched”	44 HIGH-severity OS findings remain with **no fixed version available** (§4.3).
“Cold start measured”	The container-restart measurement was blocked by the environment (§7).

The CI pipeline itself **is** verified: run #9 on `main` passed all six jobs (§3.1). That row used to read “not executed” — it was corrected after the run, not before.

Stating these up front is the point of the document. A security report that lists only what passed is marketing.

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2. Delivery pipeline design



Design decisions worth defending:

- **Deploy is `workflow_dispatch` only.** A detection platform must not reach production because someone merged a branch. The deploy job also **re-runs every gate** rather than trusting the PR run, and sits behind a GitHub `production` environment (reviewer approval required).
- **Keyless cloud auth.** `google-github-actions/auth` exchanges GitHub’s OIDC token via Workload Identity Federation. There is no service-account JSON key in the repo, in secrets, or in history — so there is nothing to leak. `permissions: id-token: write` is scoped to the deploy workflow only.
- **Secret scanning uses `fetch-depth: 0`.** A secret removed in a later commit is still in the history and still compromised. Shallow clones give a false sense of safety.
- **`--ignore-unfixed` on the Trivy gate.** Explained in §4.3: failing on findings nobody can fix trains the team to ignore the gate. Unfixable findings are still exported as SARIF.

- **A functional smoke test gates the build.** Scanning a container proves nothing about whether it runs; CI boots it and asserts that the attack simulation still produces ≥10 alerts.
- **Dependabot** covers pip, docker and actions weekly — the supply-chain half that scanners miss.

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3. What was executed, and where

Gate	Tool	Executed	Environment
Lint	ruff 0.17.1	yes — 0 findings	local
Unit tests	pytest 8.4.2	yes — 13 passed	local
SAST	bandit 1.9.4	yes	local
Dependency CVEs	pip-audit 2.10.1	yes	local
Secret scan	gitleaks (container)	yes	local
Dockerfile lint	hadolint (container)	yes	local
Image CVEs	trivy (container)	yes, before/after	local
Smoke test	curl + docker	yes, by hand (§6)	local
CI execution	GitHub Actions	yes — run #9, 6/6 jobs green	github.com/Rhack-INDIA/aegis-soc
Cloud deploy	Cloud Run	no	not run — no credentials

The binary scanners (trivy, gitleaks, hadolint) were run **as containers** rather than installed, so no privileged action was required.

3.1 The first CI runs: six failures, none of them in the application

The pipeline's first three executions failed. Documenting them matters more than the eventual green run, because every failure was a defect in the *pipeline*, not the product:

Run	Failing job	Root cause	Fix
#1	container	`aquasecurity/trivy-action@0.28.0` — **this tag does not exist** (HTTP 404). Job died at "Set up job"	Replaced with `docker run aquasec/trivy` — no version drift, no marketplace dependency
#1	secret scan	`gitleaks/gitleaks-action@v2` requires a **paid licence for organisation accounts**; `Rhack-INDIA` is an org	Replaced with the gitleaks CLI in a container (MIT, no licence)
#1	dependencies	Genuine finding: `pytest` advisory	Fixed by pinning `pytest>=9.0.3` and splitting dev deps out
#2	secret scan	**Gitleaks was right**: `GPG_KEY=…` inside committed `evals/security/trivy.sbom.json`	Artefact untracked; `.gitleaks.toml` allowlists only that published base-image value

Run	Failing job	Root cause	Fix
#2	container	`apt-get upgrade` blocked on dpkg conffile prompts in a clean CI image (`DEBIAN_FRONTEND` unset)	`DEBIAN_FRONTEND=noninteractiv` + `--force-confdef/--force-confold`
#2	container (cosmetic)	The *informational* Trivy step reported **success while the build had failed**, because `if: always() + || true` swallowed the error	`if: always()` removed — a skipped step is honest, a green step that tested nothing is not

That last one is the most transferable lesson: a passing check that executes nothing is worse than a failing one, because it converts a broken pipeline into a green dashboard. It was found by reading the job's step list rather than trusting the overall status.

Final state — run #9 on `main` (commit `8205924`):

```
SAST (bandit)                              success
container (hadolint + trivy + smoke)       success
dependencies (pip-audit)                   success
lint (ruff)                                success
secret scan (gitleaks, full history)       success
unit tests (pytest)                        success
6/6 jobs green
```

Dependabot immediately opened a PR bumping the base image to `python-3.14-slim`, confirming the supply-chain automation in `.github/dependabot.yml` is live rather than aspirational.

---

4. Security findings register

Severity uses tool output, not opinion. Status is one of **Fixed**, **Open**, **Accepted**.

4.1 Summary

#	Finding	Severity	Tool	Status
F1	`starlette 0.41.3` — DoS via Range header merging (CVE-2025-62727)	HIGH	trivy	**Fixed**
F2	`starlette 0.41.3` — SSRF / NTLM credential theft via UNC paths in StaticFiles (CVE-2026-48818)	HIGH	trivy	**Fixed**

#	Finding	Severity	Tool	Status
F3	`starlette 0.41.3` — `request.form()` limits ignored, DoS (CVE-2026-54283)	HIGH	trivy	**Fixed**
F4	44 × OS packages (`util-linux` family)	HIGH	trivy	**Accepted** — no upstream fix
F5	`pytest 8.4.2` (PYSEC-2026-1845)	MEDIUM*	pip-audit	**Fixed** — bumped to 9.0.3
F6	Dockerfile `CMD` shell form (DL3025)	MEDIUM*	hadolint	**Fixed**
F7	Dockerfile `USER` non-numeric (DL3066)	LOW*	hadolint	**Fixed**
F8	4 × `try/except` defensive blocks (B110/B112)	LOW	bandit	**Accepted**
F9	86 lint findings (24 under project policy)	INFO	ruff	**Fixed** — 0 remaining

* These tools do not publish CVSS; the rating shown is the tool's own severity/threshold class (hadolint `warning`/`info`, pip-audit advisory). Not the same scale as a CVE's CVSS.

4.2 F1–F3 — Python dependency chain (fixed)

The application pins `fastapi`, which resolves `starlette`. The image shipped **starlette 0.41.3**, which carries three HIGH advisories. Fix: bump the framework pins, not `--ignore` the finding.

```
before:  fastapi 0.115.6 → starlette 0.41.3   → Python HIGH: 3
after:   fastapi 0.146.0 → starlette 1.9.0    → Python HIGH: 0
```

Verification that the upgrade is safe is behavioural, not hopeful: **13/13 tests pass** on the new versions, including the injection-defence and policy-gate tests.

4.3 F4 — 44 HIGH OS findings with no fix available (accepted, with reasoning)

Trivy's JSON output was parsed rather than eyeballed, because the distinction that matters is **fixable vs unfixable**:

```
total HIGH/CRITICAL: 44
by class: {'os-pkgs': 44}
fixable: {}                          ← every finding has an empty FixedVersion
top packages: bsdutils(4), libblkid1(4), liblastlog2-2(4), libmount1(4),
              libsmartcols1(4), libuuid1(4)
```

All 44 are the `util-linux` family in Debian bookworm, and **none has a fixed version** at this base image. `apt-get upgrade` was already added to the build and removed nothing, because there is nothing to install. Options and the decision:

Option	Assessment
Pin `--ignore-unfixed` in the gate	**Chosen.** Fails only on actionable findings; full inventory still exported as SARIF
Switch to distroless / Alpine	Would remove the package family entirely, but changes libc and adds rebuild risk four days before a demo
Run as non-root + read-only FS	**Chosen as compensating control** — implemented: `USER 10001`, and the container needs no mount privileges
Downgrade the finding severity	Rejected — dishonest

Residual risk: the vulnerable code paths are `util-linux` mount helpers. The container runs as UID 10001 with no `CAP_SYS_ADMIN`, no host mounts, and never invokes `mount`/`su`/`login`. The realistic exploit precondition is not reachable in this deployment, which is why "Accepted" is defensible rather than convenient.

4.4 F5 — pytest advisory (fixed)

```
Found 2 known vulnerabilities in 1 package
pytest 8.4.2   PYSEC-2026-1845   fix: 9.0.3
```

Two rows are the same advisory reported per dependency path. `requirements.txt` capped pytest below the fix. Two changes resolved it, and the second was the more valuable one:

```
- pytest>=8.0,<9.0        (in requirements.txt)
+ pytest>=9.0.3,<10       (in requirements-dev.txt)
```

`pytest` is test-only tooling, so it was moved out of the runtime requirements entirely and into `requirements-dev.txt`. That removed the advisory from the shipped image's dependency surface, shrank the image by 19 MB (355 → 336 MB), and let `pip-audit` assert something meaningful about both sets: **runtime** dependencies (what an attacker can reach) and **dev** dependencies (what CI runs). The deps job now audits both, and both pass on run #9.

4.5 F6–F7 — Dockerfile hardening (fixed)

hadolint's first run:

```
-:21 DL3025 warning: Use arguments JSON notation for CMD and ENTRYPOINT arguments
```

Fixed properly rather than suppressed: a plain JSON `CMD` cannot expand Cloud Run's `$PORT`, which is precisely why the shell form was there. The fix is a tiny entrypoint that solves both:

```
#!/bin/sh
exec uvicorn app.main:app --host "${HOST:-0.0.0.0}" --port "${PORT:-8080}" --workers "${WORKE
```

```
COPY docker-entrypoint.sh /app/docker-entrypoint.sh
RUN chmod +x /app/docker-entrypoint.sh
USER 10001
CMD ["/app/docker-entrypoint.sh"]
```

Second run then flagged `DL3066` (non-numeric user), also fixed by using `USER 10001`. Final state:

```
docker run --rm -i hadolint/hadolint < Dockerfile
exit=0
```

Additional hardening applied at the same time: `apt-get upgrade` in the build, a dedicated non-root user, and `chown` of the application tree.

4.6 F8 — four low-severity bandit findings (accepted)

```
B110 try_except_pass     app/detect.py:159, app/main.py:298
B112 try_except_continue app/detect.py:221, app/store.py:113
Severity: Low (×4)   Medium: 0   High: 0
```

All four are intentional resilience boundaries, e.g. a malformed rule predicate must not take down ingestion, and one corrupt Firestore document must not abort a listing. Each is annotated in code. Policy: **CI fails on MEDIUM and above** (`bandit -ll`), so these are visible but non-blocking — the correct treatment for accepted risk, as opposed to blanket `# nosec`.

4.7 F9 — lint findings (fixed)

Initial run under ruff's default rule set: **86 findings**. A `ruff.toml` policy was committed selecting `E,F,W,I,UP,B` (correctness, imports, modernisation, bugbear) and excluding stylistic churn, which reduced the set to **24 genuine issues**:

```
F401(8) unused-import   UP017(7) timezone.utc → UTC   UP037(2) quoted annotation
I001(2) import order    UP035(1) deprecated import    UP007(1) Union → |
B008(1) call in default E741(1) ambiguous name        F841(1) unused variable
```

`ruff check --fix` was blocked by an approval timeout, so all 24 were fixed **by hand** — which is what the B008 fix required anyway: `body: Union[dict, list] = Body(...)` became `body: Annotated[dict | list, Body()]`, the FastAPI-idiomatic form that satisfies both the linter and the framework's body parsing.

Two fixes needed care: removing the `timezone` import in four modules meant every `timezone.utc` call site had to become `UTC` in the same edit, or the module would fail at import with a `NameError` rather than a lint error. Verified by re-running the corpus generator and the app, not just the linter:

```
$ ruff check app lab scripts tests
All checks passed!
```

4.8 Secret scanning

First run is instructive and is reported as a **methodological finding, not a pass**:

```
$ gitleaks detect --source=/repo
ERR [git] fatal: not a git repository
INF 0 commits scanned.
INF scanned ~0 bytes (0) in 44.9ms
INF no leaks found          ← meaningless: nothing was scanned
```

A scan of zero commits reporting "no leaks" is exactly the kind of green check that hides a compromise. After `git init` + an initial commit:

```
INF 1 commits scanned.
INF scanned ~286593 bytes (286.59 KB) in 278ms
INF no leaks found          ← now a real result
```

Verified additionally by construction: the Gemini key is read from `os.getenv` / Secret Manager and is never written to disk; `.gitignore` covers `.env`, `*.tfstate`, `*.pem`, `data/*.db`.

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5. Resource consumption (measured)

5.1 Method

- Image: `aegis-soc:fixed` (hardened), run with `-p 8098:8080`, SQLite store, `AUTO_TRIAGE=1`, single uvicorn worker, 1 vCPU container.

- Load generator: `scripts/loadtest.py` (asyncio + httpx), 20 concurrent clients, 20–30 s per mode.
- Resource sampling: `docker stats --no-stream` once per second; CPU averaged and peaked; RSS from `MemUsage`.

5.2 CPU and memory

State	CPU avg	CPU peak	RSS avg	RSS peak
Idle	**0.1%**	0.1%	**53.2 MiB**	53.2 MiB
Ingest, 20 concurrent clients, 411.7 rps	**22.6%**	**47.1%**	**53.4 MiB**	**54.3 MiB**

Two things follow directly:

1. **The 512 MiB Cloud Run limit is ~10× oversized for this workload.** 54 MiB peak supports a reduction to **256 MiB**, which doubles the memory-seconds available in the free tier.
2. **Memory is flat under load** (53.2 → 54.3 MiB, +2%), which is the signature of an allocation-stable service. There is no per-request leak in this path; the number is stable because alerts are streamed to storage rather than accumulated in memory.

5.3 Throughput and latency

Endpoint	Throughput	p50	p95	p99	Errors
`GET /api/health`	452.9 rps	26.5 ms	135.3 ms	202.8 ms	0.0%
`GET /api/alerts?limit=50`	448.7 rps	26.1 ms	135.9 ms	213.3 ms	0.0%
`POST /ingest` (rules + fusion + triage)	**411.7 rps**	23.9 ms	154.2 ms	249.8 ms	**0.0%**

The ingest row is taken from the **30-second run during which CPU and memory were sampled** (§5.2), so the throughput figure and the resource figures describe the same workload. The health and read rows are from a separate 20-second run (397.7 rps ingest, p95 160.8 ms) — directionally identical, and both raw results are kept in `evals/resource/` rather than one being quietly dropped.

Caveat that must travel with these numbers: the generator itself is the ceiling. Health-check and alert-read throughput converge at ~450 rps regardless of work done, which is the signature of client-side saturation in a single-threaded Python load generator, not server capacity. Treat ~400 rps ingest as a **floor on server capability**, not a measured maximum. A k6/Go generator is required to find the true knee.

5.4 Image size

Stage	On disk	Compressed (registry)
Pre-hardening (`aegis-soc:test`)	351 MB	**72.8 MiB** (measured)
Post-hardening (`aegis-soc:fixed`)	355 MB	not measured (blocked, §7)

Stage	On disk	Compressed (registry)
After dependency split (runtime only)	**336 MB**	not measured

The `apt-get upgrade` added 4 MB; moving `pytest` out of the runtime layer later removed 19 MB. Against Artifact Registry's 0.5 GB free allowance the compressed figure is what matters: ~73 MiB leaves room for roughly six tagged builds.

5.5 What the free tier supports

Derived from the measured ingest rate, not estimated:

Free-tier limit	At the measured rate
2,000,000 Cloud Run requests/month	≈ **81 minutes** of sustained 411 rps
360,000 GiB-seconds memory/month	≈ 190 hours at 512 MiB, ≈ 390 hours at 256 MiB
180,000 vCPU-seconds/month	≈ 50 hours of one vCPU at 100%
Firestore 20,000 writes/day	20,000 alerts/day (events are not written per-document)

This is the most useful operational number in the document: **the pipeline is cheap per request and expensive in aggregate.** A demo is free; continuous production ingest is not, which is why `--max-instances=3` and an ingest cap are load-bearing guardrails rather than decoration.

---

6. Functional verification performed by hand

Check	Result
`pytest` (13 tests)	13 passed
Container boots, `GET /api/health`	200, `"status":"ok"`
`POST /api/simulate/attack` in container	**12 alerts**
Live HTTPS endpoint (Cloudflare Tunnel)	200, ~0.68 s round trip
Injection test via API	ASR 58.3% → 0.0%
Live deployment (AWS EC2)	`http://3.106.151.93` — `/api/health` 200, full 11-rule attack chain, 14 alerts all triaged, injection ASR 0.583 → 0.0 verified from outside
`terraform/deploy.sh` syntax	`bash -n` clean

The live instance was verified from a separate host, not from itself: health, alert queue contents, statistics and the injection endpoint were all fetched over the public internet. Its header originally read "GCP free tier" while running on AWS — corrected, along with a stat card that reported the ATT&CK corpus size instead of techniques actually detected.

---

7. Not verified (and why)

Item	Blocker
Cloud Run deployment	No GCP credentials

Item	Blocker
Cold-start latency	`docker restart` measurement blocked by an approval timeout
Post-hardening compressed image size	Same blocked command sequence
pytest 9.x compatibility	`pip install -U pytest` blocked; tests are standard-API so risk is low but **unverified**

Every one of these is a five-minute manual step (§8), not a design problem — but until they run, the corresponding claim stays “not verified”.

---

8. Reproduce

```
# local gates (no sudo; scanners run as containers)
pip install ruff bandit pip-audit
./scripts/security_checks.sh

# individual numbers
ruff check app lab scripts tests
python -m pytest -q
bandit -ll -r app
pip-audit -r requirements.txt
docker run --rm -i hadolint/hadolint:latest < Dockerfile
docker run --rm -v "$PWD":/repo zricethezav/gitleaks:latest detect --source=/repo --redact

# image inventory, fixable vs not
docker run --rm -v /var/run/docker.sock:/var/run/docker.sock \
  -v "$PWD/evals/security":/out aquasec/trivy:latest image \
  --format json -o /out/trivy.json --severity HIGH,CRITICAL aegis-soc:fixed

# resource consumption
docker run -d --name aegis-bench -p 8098:8080 -e STORE=sqlite \
  -e SQLITE_PATH=/tmp/bench.db aegis-soc:fixed
python scripts/loadtest.py --url http://127.0.0.1:8098 --mode all --seconds 20 --concurrency
docker stats --no-stream aegis-bench
```

Artifacts: `evals/security/{trivy.txt,trivy_after.txt,trivy.json,gitleaks.txt,hadolint.txt,ruff.txt,bandit.txt,pip-audit.txt}`, `evals/resource/{consumption.json,loadtest.json,stats.txt}`.

---

9. Conclusion

The pipeline treats security as **gates that can fail**, not as a checklist: lint, tests, SAST, dependency audit, secret scan, Dockerfile lint, image scan with a fixability-aware threshold, and a functional smoke test, behind a manual, reviewer-approved deploy that re-runs every gate. It is live on `main` — **run #9, six of six jobs green** — and its own first three executions failed on pipeline defects (a non-existent action tag, an org-only licence gate, a dpkg prompt) rather than product defects; §3.1 records each one with the fix.

Ten findings were surfaced across two scanner rounds. Seven were fixed and verified — the three HIGH Python CVEs, both Dockerfile issues, the entire lint backlog, and the pytest advisory — with the CVE fix proven safe by the existing test suite and the dependency fix proven by a green deps job. Four OS findings were **accepted with a stated residual risk and a compensating control**. **No finding is left in an “open” state.**

The resource measurements change a design decision rather than merely documenting one: **54 MiB peak RSS under 411 rps means the 512 MiB memory request should be halved**, and the per-request profile (~400+ rps at 0.0% error) is far beyond what a coursework demo needs — which is exactly why the free tier's monthly request allowance, not compute, is the real budget constraint.